

D1: Simplified $q(\mathbf{L})$ Update Under $\eta = \mathbf{YF}\tilde{\beta}$

Cluster B (Cox-on-YF Reformulation) — Derivation D1

Cross-reference: `derivations/Modular_UpdateDerivations/ q(L)` section

Andrew Walther

2026-04-29

Contents

1	Context and Goal	2
2	ELBO Decomposition Under the New Model	2
2.1	Survival Term Does Not Involve $\mathbf{L}$	2
2.2	Genomics Likelihood Depends on $\mathbf{L}$	2
3	Factor-wise ELBO for $q(l_k)$	3
4	Variational Update for $q(l_{ik})$ — Pure-Genomics EBNM	3
5	Comparison to Original $q(L)$ Update	4
6	Code Implications	4
7	Verification Checklist	4

1 Context and Goal

In the original SSBMF model, the Cox proportional-hazards linear predictor is

$$\eta_i = \mathbf{l}_i^\top \boldsymbol{\beta}, \quad \text{where } \mathbf{l}_i = \text{row } i \text{ of the latent loading matrix } \mathbf{L}. \quad (1)$$

This makes $\mathbf{L}$ appear in *both* the genomics likelihood ($\mathbf{Y} \approx \mathbf{L}\mathbf{F}^\top$) and the Cox likelihood (through η_i), so the variational update for $q(\mathbf{L})$ must balance two information sources (“dual-source EBNM”).

Cluster B reformulation. Replace the linear predictor with

$$\eta_i = (\mathbf{y}_i \mathbf{F}) \tilde{\boldsymbol{\beta}} = \sum_k \left(\sum_j y_{ij} f_{jk} \right) \tilde{\beta}_k, \quad (2)$$

where $\tilde{\boldsymbol{\beta}}$ is the CAVI variational parameter and $\mathbf{Z}_F = \mathbf{Y}\mathbf{F} \in \mathbb{R}^{n \times K}$ are *observed* projection scores (held fixed per outer iteration).

Goal of D1. Show that under (2), $\mathbf{L}$ no longer appears in the Cox likelihood term, so the ELBO’s dependence on $q(\mathbf{L})$ reduces to the genomics likelihood alone, yielding a pure-genomics EBNM update.

2 ELBO Decomposition Under the New Model

The full ELBO is

$$\mathcal{L} = \underbrace{\mathbb{E}_q[\log p(\mathbf{Y} \mid \mathbf{L}, \mathbf{F}, \boldsymbol{\tau})]}_{\text{genomics}} + \underbrace{\mathbb{E}_q[\log p(\mathbf{t}, \boldsymbol{\delta} \mid \mathbf{Y}, \mathbf{F}, \tilde{\boldsymbol{\beta}})]}_{\text{survival (new)}} - \text{KL}[q(\mathbf{L}) \parallel p(\mathbf{L})] - \text{KL}[q(\mathbf{F}) \parallel p(\mathbf{F})] - \text{KL}[q(\tilde{\boldsymbol{\beta}}) \parallel p(\tilde{\boldsymbol{\beta}})] - \text{KL}[q(\boldsymbol{\tau}) \parallel p(\boldsymbol{\tau})] \quad (3)$$

Key observation

Under (2), the survival term $\mathbb{E}_q[\log p(\mathbf{t}, \boldsymbol{\delta} \mid \mathbf{Y}, \mathbf{F}, \tilde{\boldsymbol{\beta}})]$ depends on $\mathbf{Z}_F = \mathbf{Y}\mathbf{F}$ and $\tilde{\boldsymbol{\beta}}$, but **not on $\mathbf{L}$** . The matrix $\mathbf{L}$ appears only in the genomics likelihood.

2.1 Survival Term Does Not Involve $\mathbf{L}$

The Cox partial likelihood (and its Taylor-linearised surrogate used in CAVI) depends on $\eta_i = \sum_k (z_F)_{ik} \tilde{\beta}_k$ where $(z_F)_{ik} = \sum_j y_{ij} f_{jk}$. The terms $(z_F)_{ik}$ are functions of the *observed* $\mathbf{Y}$ and the *factor* $\mathbf{F}$; they do not involve the loadings $\mathbf{L}$.

Therefore

$$\frac{\partial}{\partial q(\mathbf{L})} \mathbb{E}_q[\log p(\mathbf{t}, \boldsymbol{\delta} \mid \mathbf{Y}, \mathbf{F}, \tilde{\boldsymbol{\beta}})] = 0. \quad (4)$$

2.2 Genomics Likelihood Depends on $\mathbf{L}$

The genomics model is $\mathbf{Y} \approx \mathbf{L}\mathbf{F}^\top + \mathbf{E}$ where $e_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_j^{-1})$. The expected log-likelihood is

$$\mathbb{E}_q[\log p(\mathbf{Y} \mid \mathbf{L}, \mathbf{F}, \boldsymbol{\tau})] = -\frac{1}{2} \sum_{i,j} \tau_j \mathbb{E}_q[(y_{ij} - \mathbf{l}_i^\top \mathbf{f}_j)^2]. \quad (5)$$

This depends on $q(\mathbf{L})$ through $\mathbb{E}_q[\mathbf{l}_i]$ and $\mathbb{E}_q[\mathbf{l}_{ik}^2]$.

3 Factor-wise ELBO for $q(l_k)$

In Gauss-Seidel CAVI, we update $q(\mathbf{l}_k)$ holding all other factors fixed. Define the partial residual

$$R_k = \mathbf{Y} - \sum_{k' \neq k} \mathbb{E}_q[\mathbf{l}_{k'}] \mathbb{E}_q[\mathbf{f}_{k'}]^\top \in \mathbb{R}^{n \times p}. \quad (6)$$

Isolating factor k in (5):

$$\begin{aligned} \mathbb{E}_q[\log p(\mathbf{Y} \mid \mathbf{L}, \mathbf{F}, \boldsymbol{\tau})] \Big|_k &= -\frac{1}{2} \sum_{i,j} \tau_j \mathbb{E}_q[(r_{k,ij} - l_{ik} f_{jk})^2] + \text{const} \\ &= -\frac{1}{2} \sum_i \left[\mathbb{E}_q[l_{ik}^2] \sum_j \tau_j \mathbb{E}_q[f_{jk}^2] - 2 \mathbb{E}_q[l_{ik}] \sum_j \tau_j \mathbb{E}_q[r_{k,ij}] \mathbb{E}_q[f_{jk}] \right] + \text{const}. \end{aligned} \quad (7)$$

Survival terms: absent

The factor- k contribution to the ELBO that involves $q(l_{ik})$ is given entirely by (7). There are **no** w_i , $\tilde{\beta}_k$, $z_{\text{no-}k}$ or other survival terms present, because $\mathbf{L}$ is absent from the survival likelihood under the Cluster B reformulation.

4 Variational Update for $q(l_{ik})$ — Pure-Genomics EBNM

Completing the square in (7), the optimal $q(l_{ik})$ is the EBNM solution with pseudo-observation

$$x_{L,ik} = \frac{B_{L,ik}}{A_{L,i}}, \quad s_{L,ik} = \frac{1}{\sqrt{A_{L,i}}}, \quad (8)$$

where

$$A_{L,i} = \sum_j \tau_j \mathbb{E}_q[f_{jk}^2], \quad (9)$$

$$B_{L,ik} = \sum_j \tau_j \mathbb{E}_q[r_{k,ij}] \mathbb{E}_q[f_{jk}]. \quad (10)$$

Note that $A_{L,i}$ is the *same* for all patients i (it sums over features j), so the subscript i is redundant in practice — written here for clarity.

EBNM call for $q(l_k)$ under Cluster B

Input: $x_{L,k} = B_{L,k}/A_L$ (n-vector), $s_L = 1/\sqrt{A_L}$ (scalar, same for all i)

Prior family: point-exponential (or as configured in `prior_LF`)

Output: $\mathbb{E}_q[l_{ik}]$, $\mathbb{E}_q[l_{ik}^2]$, `ebnm_log_likelihood`

5 Comparison to Original $q(L)$ Update

Quantity	Original ($\eta = \mathbf{L}\beta$)	Cluster B ($\eta = \mathbf{Z}_F\tilde{\beta}$)
$A_{L,i}$	$(1 - \alpha) A_{\text{gen}} + \alpha \lambda w_i \mathbb{E}[\beta_k^2]$	$A_{\text{gen}} = \sum_j \tau_j \mathbb{E}[f_{jk}^2]$
$B_{L,ik}$	$(1 - \alpha) B_{\text{gen},ik} + \alpha \lambda w_i z_{\text{no-}k,i} \mathbb{E}[\beta_k]$	$B_{\text{gen},ik}$ only
Survival args	$w, \mathbb{E}[\beta_k], \mathbb{E}[\beta_k^2], z_{\text{no-}k}, \lambda, \alpha$	<i>none</i>

The Cluster B update is strictly simpler: α no longer appears (or equivalently, the $(1 - \alpha)$ factor in front of A_{gen} becomes 1, so we may drop it and set the floor directly). The parameters $w, \tilde{\beta}_k, \mathbb{E}[\tilde{\beta}_k^2], z_{\text{no-}k}, \lambda$, and `normalize_AB` are all removed from `update_L_k()`.

6 Code Implications

`code/update_L.R` — `update_L_k()`

- **Remove arguments:** `w, EBeta_k, EBeta2_k, z_no_k, lambda, normalize_AB, alpha`
- **Remove:** `A_surv` and `B_surv` computation blocks
- **Simplify:** `A_L <- pmax(A_gen, A_floor), B_L <- B_gen`
- **New signature:** `update_L_k(Tau, EF_k, EF2_k, R_k, prior_family, A_floor)`

`code/update_L.R` — `update_L_all()` Drop `w, z, EBeta, EBeta2, lambda, normalize_AB` from the outer wrapper. The Gauss-Seidel k -loop structure is preserved; only the argument threading changes.

7 Verification Checklist

D1 Verification

1. $\eta_i = \sum_k (z_F)_{ik} \tilde{\beta}_k$ contains no l_{ik} . ✓
2. The Cox Taylor expansion — $\log \text{PL} \approx \log \text{PL}_0 + \sum_i u_i \delta \eta_i - \frac{1}{2} \sum_i w_i (\delta \eta_i)^2$ — involves $\mathbf{Z}_F$ and $\tilde{\beta}$, not $\mathbf{L}$. ✓
3. Factor-wise ELBO for $q(l_k)$ contains only the genomics terms (9)–(10). ✓
4. All survival-specific arguments removed from `update_L_k()`. (to implement)
5. 171/171 tests pass after simplification. (to implement)